# Evaluating Claim-Level Verification for Small Retrieval-Augmented LLMs

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**Abstract.** Retrieval-augmented generation (RAG) reduces hallucination, yet small open models still give confident wrong answers when the retrieved evidence is incomplete. A popular remedy is a post-hoc check: split the answer into atomic claims, test each claim against the evidence, repair what fails, and abstain when the answer cannot be supported. This study asks whether this machinery earns its cost when the whole system must run on one 8 GB consumer GPU. The proposed pipeline, RAG-Verify, combines a question-aware LLM check of the answer claim with natural language inference on the remaining claims, followed by targeted revision and an abstention gate. It is compared with matched baselines on three disjoint 300-question test blocks of HotpotQA and TriviaQA using 4-bit Qwen2.5-7B and Llama-3.1-8B, with the design fixed beforehand on development data. With Qwen2.5-7B, RAG-Verify cuts the hallucination rate of standard RAG from 50.6% to 21.0% on HotpotQA and from 23.8% to 7.9% on TriviaQA. However, a simpler baseline that re-reads the evidence and revises its answer once, with no verifier, is significantly better with Qwen and statistically tied with Llama, using fewer model calls. The verifier flags 39% of initially correct HotpotQA answers, and most flagged drafts end in abstention. At this scale, the benefit attributed to claim-level verification comes mainly from a second, evidence-focused pass.

**Keywords:** Large language models · Hallucination · Retrieval-augmented generation · Fact verification · Natural language inference · Selective question answering

## 1 Introduction

Large language models produce fluent text that is not always true, a failure usually called hallucination [6]. Retrieval-augmented generation, which grounds the model in retrieved passages [10], reduces the problem but does not remove it: the retriever can miss the decisive passage, and the model may ignore or misread evidence that is present [14]. Post-hoc methods therefore inspect the output after generation.

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Most studies use large proprietary models, yet many deployments must run locally on one consumer GPU. There every extra call costs latency, and decomposition, entailment checking and revision are themselves done by small models that make mistakes. This paper tests whether stacking these imperfect steps helps. Our contributions are:

1. *RAG-Verify*, a local claim-level pipeline within 8 GB of GPU memory, with a *hybrid* verifier: a question-aware LLM check for the answer claim and an NLI model for the supporting claims.
2. A pre-registered protocol: the design is chosen on a development block by a rule fixed in advance, and all systems run once on three disjoint test blocks per dataset.
3. A negative but useful finding: self-revision with no verifier is significantly better than claim-level verification with Qwen2.5-7B and statistically tied with Llama-3.1-8B, with no more than 60% of its model calls, because verifier false alarms turn correct answers into abstentions.

Table 1 lists the abbreviations used in the paper.

**Table 1.** Abbreviations used in this paper

| Abbr. | Meaning | Abbr. | Meaning |
|---|---|---|---|
| AUROC | Area under the receiver operating characteristic curve | NLI | Natural language inference |
| CI | Confidence interval | Q4_K_M | 4-bit medium k-quant format of llama.cpp |
| CoVe | Chain-of-Verification | RAG | Retrieval-augmented generation |
| F1 | Harmonic mean of precision and recall | RARR | Retrofit Attribution using Research and Revision |
| GPU | Graphics processing unit | SAFE | Search-Augmented Factuality Evaluator |
| LLM | Large language model | SQuAD | Stanford Question Answering Dataset |
| NF4 | 4-bit NormalFloat quantization | | |

## 2 Related Work

RAG conditions generation on retrieved passages [10], and Self-RAG [1] trains the generator to emit reflection tokens that rate each passage's relevance and support and the usefulness of the output. CoVe [2] drafts an answer, plans verification questions, answers them independently, and writes a final response; RARR [4] retrieves evidence for an existing output and edits unsupported spans. Claim-level pipelines such as FActScore [13] and SAFE [18] split text into atomic

facts and check each one against a knowledge source; more recent work separates retrieval, verification and calibration into modules [3], or verifies entities against evidence [7]. MiniCheck [17] and NLI models trained on many entailment datasets [9] make claim checking affordable. Without retrieval, SelfCheckGPT [12] detects hallucinations from disagreement between samples. Declining to answer is a legitimate outcome when the model is unsure [19]; we therefore score systems by effective reliability [20], defined in Sect. 4. Our study differs from this work in scope rather than mechanism: one consumer GPU, the simple baselines a deployer would try first, and the contribution of each component.

## 3 Method

Given a question $q$, a retriever $R$ returns the top-$k$ passages $P = \{p_1, \ldots, p_k\}$, and the generator LLM $G$ produces a short response $r_0$ (answer line plus one-sentence explanation). RAG-Verify splits $r_0$ into $m$ atomic claims $C = (c_1, \ldots, c_m)$, where the *answer claim* $c_1$ states the answer, and gives each claim $c_i$ a label $\ell_i \in \{\text{SUPPORTED}, \text{CONTRADICTED}, \text{UNSUPPORTED}\}$; $\ell = (\ell_1, \ldots, \ell_m)$. If some claim is not supported, $G$ writes a revised response $r_1$. Two parameters steer the decisions: the entailment threshold $\tau$ of the NLI model $N$, and the minimum fraction $\rho$ of supported claims that an answer needs. Algorithm 1 gives the steps.

**Algorithm 1** RAG-Verify (one question)

**Require:** $q$, $R$, $G$, $N$, $\tau$, $\rho$
$P \leftarrow R(q, k)$; $\quad r_0 \leftarrow G(\textsc{Answer}(q, P))$
$C \leftarrow \textsc{Decompose}_G(q, r_0)$ {answer claim first}
$\ell \leftarrow \textsc{Verify}(q, C, P)$ {hybrid, Eq. (1)}
$r_1 \leftarrow r_0$
**if** some $\ell_i \neq$ SUPPORTED **then**
  $r_1 \leftarrow G(\textsc{Revise}(q, P, r_0, \{(c_i, \ell_i)\}))$
  **if** $r_1$ is a refusal ("I don't know") **then**
    **return** ABSTAIN
  **end if**
  $C \leftarrow \textsc{Decompose}_G(q, r_1)$; $\quad \ell \leftarrow \textsc{Verify}(q, C, P)$
**end if**
**if** $\ell_1 \neq$ SUPPORTED **or** $s(\ell) < \rho$ **then**
  **return** ABSTAIN
**end if**
**return** the answer in $r_1$

**Decomposition.** $G$ is prompted with two worked examples to rewrite the response as at most twelve self-contained claims. We move the claim containing the answer string to the front, or insert a template claim when none contains it.

**Hybrid Verification.** For a supporting claim $c_i$ ($i > 1$), $N$ returns entailment and contradiction probabilities for each premise; premises are the $k$ passages plus the concatenation of the top three, which allows two-hop facts to be entailed. With $e_i$ and $x_i$ the highest entailment and contradiction probabilities over all premises, the label is

$$\ell_i = \begin{cases} \text{SUPPORTED} & e_i \geq \tau \\ \text{CONTRADICTED} & e_i < \tau,\ x_i \geq \tau \\ \text{UNSUPPORTED} & \text{otherwise.} \end{cases} \tag{1}$$

For the answer claim $c_1$ we do not use NLI: on the development block it entailed claims such as "the answer is $X$" whenever $X$ appeared in a passage, catching only 3 of 17 wrong answers in a manual check, against 7 of 17 for an LLM judge. The answer claim is therefore judged by $G$ itself, with a prompt that asks whether the proposed answer is correct *for this question* given the passages.

**Revision and Abstention.** Flagged claims and their labels are shown to $G$ with the passages and the original response; $G$ may correct the answer or reply that it does not know. After re-verification, the system abstains if the answer claim is not supported or if the supported fraction $s(\ell) = \frac{1}{m}\,|\{i : \ell_i = \text{SUPPORTED}\}|$ falls below $\rho$.

**Design Selection.** Before any test run we fixed the rule: maximize mean effective reliability over two 100-question development blocks, breaking ties within 0.01 by coverage, then model calls. We compared three verifiers (NLI only, hybrid, LLM judge for every claim), $\tau \in \{0.3, 0.5, 0.7, 0.9\}$, and $\rho \in [0, 1]$. It selected the hybrid verifier with $\tau = 0.3$ and $\rho = 0$, i.e., abstention is driven by the answer claim alone (development reliability 0.200). This frozen configuration is reported as RAG-Verify.

## 4 Experimental Setup

**Data.** We use HotpotQA (distractor setting, validation split) [22] and the Wikipedia subset of TriviaQA (validation split) [8]. A fixed permutation (seed 2026) gives three disjoint 300-question test blocks and one development block per dataset. Corpora are built from the provided contexts: 66,635 HotpotQA paragraphs and 135,517 100-word TriviaQA passages. These are closed corpora, not full Wikipedia, so absolute scores are not comparable to open-domain results. To test the verifiers in isolation we also use 1,000 questions of HaluEval-QA [11], each with its gold knowledge snippet and one correct and one hallucinated answer.

**Models.** The generators are Qwen2.5-7B-Instruct [15] and Llama-3.1-8B-Instruct [5], both Q4_K_M, served by Ollama with temperature 0 and a fixed

seed. The retriever is bge-small-en-v1.5 [21] with exact search, $k = 5$. The NLI model is the DeBERTa-v3-large checkpoint of Laurer et al. [9]. Self-RAG uses the authors' 7B checkpoint quantized to 4-bit NF4. Hardware is a laptop with an RTX 4060 (8 GB).

**Systems.** All retrieval-based systems share the same retriever, passages, and answer format. *Closed-book* answers without passages. *Standard RAG* answers from the passages. *RAG + abstain prompt* adds an instruction to reply "I don't know" when the passages are insufficient. *RAG + self-revision* (RAG-Verify without the verifier) shows the model its first response and the passages and asks it to correct any error or abstain. *CoVe* uses the factored variant with up to five verification questions, closed-book or with retrieval. *Self-RAG* follows the authors' short-form inference.

**Metrics.** An answer is correct if, after SQuAD normalization [16], it equals or contains a gold alias. Let $A$ be the fraction of questions answered correctly and $H$ the hallucination rate (fraction answered incorrectly); abstentions count towards neither. Effective reliability with unit cost is $\Phi_1 = A - H$, which rewards a correct answer, penalizes a wrong one, and gives zero to an abstention. We report mean $\pm$ standard deviation over the three test blocks, and paired bootstrap 95% CIs (10,000 resamples) over the 900 pooled questions. On the first test block, claim precision is also measured: Llama-3.1-8B extracts claims from answered outputs and MiniCheck [17] judges each against gold evidence. Latency is the median time per question on the first 50 development questions.

## 5 Results

**Retrieval Matters Most.** Table 2 lists all systems and Table 3 the paired comparisons. With Qwen2.5-7B, retrieval raises accuracy from 21.9% to 47.4% on HotpotQA and from 57.2% to 75.6% on TriviaQA. Closed-book CoVe does not improve on the closed-book model (21.8% and 60.7%), and CoVe with retrieval is close to standard RAG ($\Phi_1$ −2.0 vs. −3.1 on HotpotQA, 54.8 vs. 51.8 on TriviaQA) at about 7.7 model calls per question.

**RAG-Verify Trades Coverage for Precision.** RAG-Verify lowers the hallucination rate of standard RAG from 50.6% to 21.0% on HotpotQA and from 23.8% to 7.9% on TriviaQA, the lowest of all systems. It abstains on 49.2% and 27.4% of questions, so accuracy falls to 29.8% and 64.7%. The net effect on $\Phi_1$ is positive and significant: +11.9 and +5.0 points. When it does answer, RAG-Verify is right 58.5% and 89.1% of the time, against 48.4% and 76.1% for standard RAG.

**Table 2.** Test results (%, mean ± standard deviation over three disjoint 300-question blocks). Acc: accuracy; Abst: abstention rate; Hall: answered and wrong; $\Phi_1$: effective reliability (higher is better)

| | HotpotQA | | | | TriviaQA | | | |
|---|---|---|---|---|---|---|---|---|
| System | Acc↑ | Abst | Hall↓ | $\Phi_1$ ↑ | Acc↑ | Abst | Hall↓ | $\Phi_1$ ↑ |
| *Backbone: Qwen2.5-7B* | | | | | | | | |
| Closed-book | 21.9±0.8 | 1.8±1.4 | 76.3±1.5 | −54.4±2.0 | 57.2±4.5 | 0.0±0.0 | 42.8±4.5 | 14.4±9.1 |
| Standard RAG | 47.4±3.9 | 2.0±0.6 | 50.6±4.2 | −3.1±8.1 | 75.6±3.3 | 0.7±0.9 | 23.8±4.2 | 51.8±7.5 |
| RAG + abstain prompt | 39.0±3.4 | 34.0±1.2 | 27.0±2.2 | 12.0±5.6 | 66.7±3.2 | 22.9±0.7 | 10.4±2.7 | 56.2±5.9 |
| RAG + self-revision | 40.6±5.5 | 34.8±0.7 | 24.7±4.8 | 15.9±10.3 | 68.6±4.0 | 22.8±1.3 | 8.7±2.7 | 59.9±6.7 |
| CoVe (closed-book) | 21.8±1.3 | 1.2±1.0 | 77.0±0.6 | −55.2±1.8 | 60.7±4.3 | 0.0±0.0 | 39.3±4.3 | 21.3±8.7 |
| CoVe + RAG | 48.3±3.8 | 1.3±0.3 | 50.3±4.0 | −2.0±7.8 | 77.0±1.7 | 0.8±0.7 | 22.2±2.3 | 54.8±4.0 |
| RAG-Verify **(ours)** | 29.8±4.2 | 49.2±2.0 | 21.0±2.2 | 8.8±6.4 | 64.7±3.3 | 27.4±1.0 | 7.9±2.5 | 56.8±5.7 |
| *Backbone: Llama-3.1-8B* | | | | | | | | |
| Closed-book | 23.3±2.0 | 1.2±0.2 | 75.4±2.1 | −52.1±4.2 | 74.9±2.0 | 0.0±0.0 | 25.1±2.0 | 49.8±4.1 |
| Standard RAG | 48.3±3.4 | 4.0±0.6 | 47.7±3.0 | 0.7±6.3 | 77.4±0.7 | 2.6±0.7 | 20.0±1.2 | 57.4±1.8 |
| RAG + abstain prompt | 46.4±2.5 | 13.9±1.4 | 39.7±2.3 | 6.8±4.7 | 74.7±0.9 | 9.9±0.8 | 15.4±1.7 | 59.2±2.5 |
| RAG + self-revision | 46.2±4.1 | 14.0±2.0 | 39.8±2.2 | 6.4±6.2 | 76.3±1.2 | 8.8±1.2 | 14.9±2.3 | 61.4±3.5 |
| RAG-Verify **(ours)** | 39.6±4.8 | 29.0±2.0 | 31.4±2.8 | 8.1±7.7 | 73.3±1.2 | 15.1±1.2 | 11.6±2.3 | 61.8±3.5 |
| *Backbone: Llama-2-7B (NF4)* | | | | | | | | |
| Self-RAG | 39.8±1.6 | 0.8±0.8 | 59.4±1.7 | −19.7±3.2 | 73.8±1.7 | 0.6±0.4 | 25.7±1.8 | 48.1±3.4 |

**Simple Baselines Are as Good or Better.** Fig. 1 compares $\Phi_1$ across systems. A one-line abstention instruction already reaches 12.0 and 56.2, not significantly different from RAG-Verify. Unconditional self-revision does better still: 15.9 and 59.9, significantly above both RAG-Verify (−7.1 and −3.1 points for RAG-Verify) and the abstention prompt (+3.9 and +3.7). It makes exactly two model calls per question, whereas RAG-Verify makes 3.7 and 3.4 on average plus NLI inference.

**Backbone.** With Llama-3.1-8B the closed-book model is much stronger on TriviaQA (74.9% vs. 57.2%), so retrieval adds less (77.4%), and the abstention

**Table 3.** Paired bootstrap differences in $\Phi_1$ (percentage points, 95% CI, 900 questions per dataset). (Q): Qwen2.5-7B backbone; (L): Llama-3.1-8B backbone; $^*$: CI excludes zero

| Comparison | HotpotQA | TriviaQA |
|---|---|---|
| RAG-Verify (Q) vs. RAG (Q) | +11.9 [+7.3, +16.6]$^*$ | +5.0 [+1.3, +8.6]$^*$ |
| RAG-Verify (Q) vs. Abstain prompt (Q) | −3.2 [−7.1, +0.7] | +0.6 [−2.1, +3.2] |
| RAG-Verify (Q) vs. Self-revision (Q) | −7.1 [−10.6, −3.8]$^*$ | −3.1 [−5.3, −0.9]$^*$ |
| RAG-Verify (Q) vs. CoVe+RAG (Q) | +10.8 [+5.8, +15.9]$^*$ | +2.0 [−1.7, +5.7] |
| RAG-Verify (Q) vs. Closed-book (Q) | +63.2 [+57.3, +69.1]$^*$ | +42.3 [+35.8, +48.7]$^*$ |
| Self-revision (Q) vs. Abstain prompt (Q) | +3.9 [+0.3, +7.4]$^*$ | +3.7 [+1.2, +6.1]$^*$ |
| RAG-Verify (L) vs. RAG (L) | +7.4 [+3.4, +11.4]$^*$ | +4.3 [+1.6, +7.1]$^*$ |
| RAG-Verify (L) vs. Abstain prompt (L) | +1.3 [−2.7, +5.3] | +2.6 [−0.3, +5.4] |
| RAG-Verify (L) vs. Self-revision (L) | +1.7 [−1.4, +4.8] | +0.3 [−1.9, +2.4] |

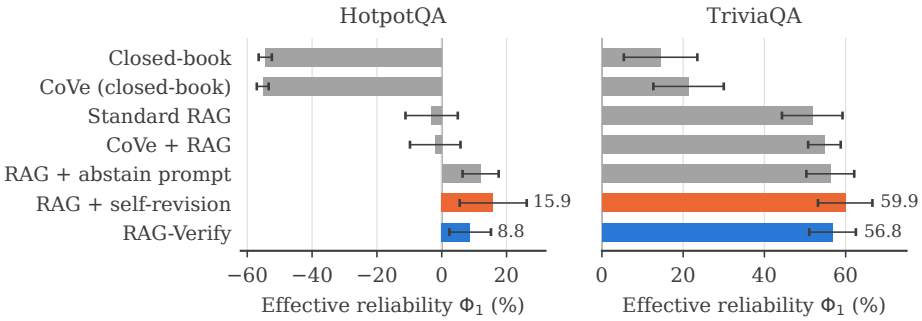



**Fig. 1.** Effective reliability $\Phi_1$ of the Qwen2.5-7B systems (mean ± standard deviation over the three test blocks); self-revision and RAG-Verify are highlighted

prompt is followed less often (13.9% and 9.9% abstention vs. 34.0% and 22.9% for Qwen). RAG-Verify again beats standard RAG significantly (+7.4 and +4.3 points) with the lowest hallucination rate among Llama systems (31.4% and 11.6%). Unlike with Qwen, it is statistically tied with self-revision (8.1 vs. 6.4, 61.8 vs. 61.4) and the abstention prompt, but needs 5.2 and 4.9 model calls against two. Self-RAG answers almost every question (0.8% and 0.6% abstention) with the highest hallucination rate of the retrieval-based systems (59.4% and 25.7%; $\Phi_1$ −19.7 and 48.1); this comparison is only indicative, as its backbone and quantization differ from ours.

## 6 Analysis

**Table 4.** Breakdown of RAG-Verify's decisions, computed from the saved traces (Qwen2.5-7B, 900 test questions per dataset)

| | HotpotQA | TriviaQA |
|---|---|---|
| First draft correct | 425 | 682 |
| First draft flagged by verifier | 466 | 282 |
| of which first draft was correct | 164 | 134 |
| Abstained: model refused in first draft | 19 | 7 |
| Abstained: model refused after revision | 380 | 233 |
| Abstained: final gate (answer claim) | 44 | 7 |
| of which withheld answer was correct | 24 | 3 |

**False Alarms Push the Model Towards Refusal.** Table 4 explains the gap to self-revision. On HotpotQA the verifier flags 466 of 900 first drafts, sending

back 38.6% of correct first answers (19.6% on TriviaQA). Most abstentions (380 of 443 on HotpotQA, 233 of 247 on TriviaQA) are the model itself answering "I don't know" after being told that some claims are unsupported: 82% of flagged HotpotQA drafts (380 of 466) and 83% of flagged TriviaQA drafts (233 of 282) end in a refusal. The final gate adds few abstentions, and on HotpotQA more than half of them (24 of 44) withhold a correct answer. Self-revision re-checks every answer without signaling an error, so it keeps more correct answers. On the development block, the hybrid verifier and self-revision did not differ significantly ($-0.025$, 95% CI $[-0.09, 0.04]$).

**Table 5.** Ablations on the first test block only (300 questions per dataset, Qwen2.5-7B). $\Phi_1$ in %; $\Delta$: paired difference from full RAG-Verify with 95% bootstrap CI. $^*$: CI excludes zero

| | HotpotQA | | TriviaQA | |
|---|---|---|---|---|
| Variant | $\Phi_1$ | $\Delta$ [95% CI] | $\Phi_1$ | $\Delta$ [95% CI] |
| RAG-Verify (full, $k$=5, $\tau$=0.3) | 15.7 | – | 55.7 | – |
| − retrieval (sample consistency) | −9.7 | −25.3 [−35.0, −15.7]$^*$ | 36.0 | −19.7 [−28.7, −10.3]$^*$ |
| − claim decomposition | 15.3 | −0.3 [−4.0, +3.3] | 54.7 | −1.0 [−4.0, +1.7] |
| − verifier (self-revision) | 26.0 | +10.3 [+4.0, +16.7]$^*$ | 58.0 | +2.3 [−1.3, +6.0] |
| − revision | 15.3 | −0.3 [−3.7, +3.3] | 55.0 | −0.7 [−3.7, +2.0] |
| − abstention gate | −3.7 | −19.3 [−27.0, −11.7]$^*$ | 46.3 | −9.3 [−15.3, −3.3]$^*$ |
| Verifier: NLI only | 9.7 | −6.0 [−12.7, +0.7] | 51.3 | −4.3 [−8.7, +0.0] |
| Verifier: LLM judge only | 13.0 | −2.7 [−8.7, +3.3] | 52.0 | −3.7 [−7.7, +0.0] |

**Ablations.** Table 5 removes one component at a time. Without retrieval, where claims are checked by agreement between five sampled answers as in SelfCheckGPT, $\Phi_1$ falls by 25.3 and 19.7 points; without the abstention gate the hallucination rate returns to that of standard RAG (50.0% vs. 47.3% on HotpotQA, 25.7% for both on TriviaQA). Removing claim decomposition or revision changes $\Phi_1$ by at most one point. Removing the verifier alone gives a significant gain (+10.3 points on HotpotQA). As sensitivity checks (full grid in the repository), $\tau$ matters only at 0.9 on TriviaQA ($-4.0$); $k = 1$ hurts RAG-Verify ($-15.0$ and $-14.3$), whereas $k = 3$ or 10 makes no significant difference.

**Error Analysis.** A *lost answer* is an abstention after a correct first draft: there are 156 on HotpotQA and 100 on TriviaQA, and the answer-claim judge rejected the correct answer in 149 and 87 of them. Of the 189 and 71 wrong answers, 60 and 22 had no gold answer string in the retrieved passages, and in 82 and 37 the judge accepted a wrong answer although the gold answer had been retrieved. In a sample of 25 lost and 25 wrong answers (one reading, no second annotator), 9 were rejected although the passages stated the answer, and 9 named a related

entity instead of the attribute asked for. The judge thus checks whether an answer is mentioned and plausible, not whether it fills the slot asked for.

**Verifiers in Isolation.** On HaluEval-QA, with the gold knowledge as evidence, NLI ranks hallucinated above correct answers with an AUROC of 0.876 (95% CI 0.860–0.890). At $\tau = 0.3$, however, it flags only 607 of the 1,000 hallucinated answers, with 40 false alarms on correct ones. The LLM judge is more balanced (accuracy 0.810, F1 0.805 vs. 0.737 for NLI) but flags 164 of the 1,000 correct answers despite clean evidence.

**Claim Precision and Cost.** On the first test block, 72.5% and 71.2% of the claims in RAG-Verify's answered outputs are supported by gold evidence, against 62.0% and 61.5% for standard RAG and 68.5% and 67.7% for self-revision; precision covers answered questions only, so systems that abstain more are measured on easier subsets. The median time per question is 3.7 s and 5.2 s for RAG-Verify, 2.0 s for self-revision, 1.1 s for standard RAG, and 6.9 s and 7.6 s for CoVe with retrieval.

**Discussion and Limitations.** Claim-level verification suits settings where a wrong answer costs far more than a refusal; under a symmetric cost, one unguided revision pass obtains most of its benefit. The evidence is limited to two short-answer benchmarks, closed corpora and string-match scoring. The quantized server is not bit-reproducible, though repeated runs left final answers unchanged, and the claim-precision judge was not validated against human judgements on these data.

## 7 Conclusion

A fully local claim-level verify-and-revise pipeline reliably reduces hallucinations relative to standard RAG, but unconditional self-revision is as good in effective reliability (Llama-3.1-8B) or significantly better (Qwen2.5-7B), and cheaper. For small open models, the second look at the evidence matters more than the fine-grained verdicts meant to guide it. Future work should test whether calibrated verifier scores can recover the discarded correct answers.

**Code Availability and Use of AI Assistance.** Code, prompts, outputs and logs: https://github.com/shivamsengar135/RAG-Verify. An AI assistant (Claude, Anthropic) helped write the code and draft and edit the text; all reported numbers come from the released code.

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